



Cambridge IGCSE™

CANDIDATE
NAME

CENTRE
NUMBER

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NUMBER

PHYSICS

0625/62

Paper 6 Alternative to Practical

February/March 2026

1 hour

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

1 A student determines the density of a glass ball.

(a) The student:

- pours water into a measuring cylinder until it is approximately half full
- places the measuring cylinder of water on a top-pan balance
- records the combined mass m_1 of the measuring cylinder and water.

$$m_1 = \dots\dots\dots 125.6 \text{ g}$$

(i) Describe **one** technique for using a top-pan balance to measure the mass of the measuring cylinder and water to ensure an accurate reading.

.....
 [1]

(ii) Figure 1.1 shows the water level in the measuring cylinder.

Record the volume V_1 of water in the measuring cylinder.

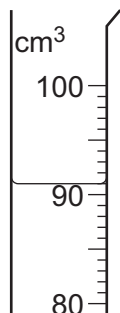


Figure 1.1

$$V_1 = \dots\dots\dots \text{ cm}^3 \text{ [1]}$$

(iii) Describe **two** techniques that the student uses when measuring the volume of the water to ensure an accurate reading.

1

 2
 [2]

- (b) The student places four glass balls into the water in the measuring cylinder, ensuring no water splashes out.

The student records the mass m_2 of the measuring cylinder, water and 4 glass balls.

$$m_2 = \dots\dots\dots 132.8\text{g}$$

- (i) Calculate the mass m of one glass ball.

$$m = \dots\dots\dots \text{g} \quad [1]$$

- (ii) The student records the volume V_2 of the water and four glass balls.

$$V_2 = \dots\dots\dots 94\text{cm}^3$$

Calculate the volume V of one glass ball.

$$V = \dots\dots\dots \text{cm}^3 \quad [1]$$

- (iii) Calculate the density ρ of a glass ball. Use your values from **1(b)(i)** and **1(b)(ii)** and give the unit. Use the equation shown.

$$\rho = \frac{m}{V}$$

$$\rho = \dots\dots\dots \text{unit} \dots\dots\dots [2]$$

- (c) Suggest **two** improvements that will help a student to determine a more accurate value for the volume of a glass ball.

1

.....

2

.....

[2]

[Total: 10]

[Turn over]



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- 2 A student investigates the behaviour of a spring.

Refer to Figure 2.1.

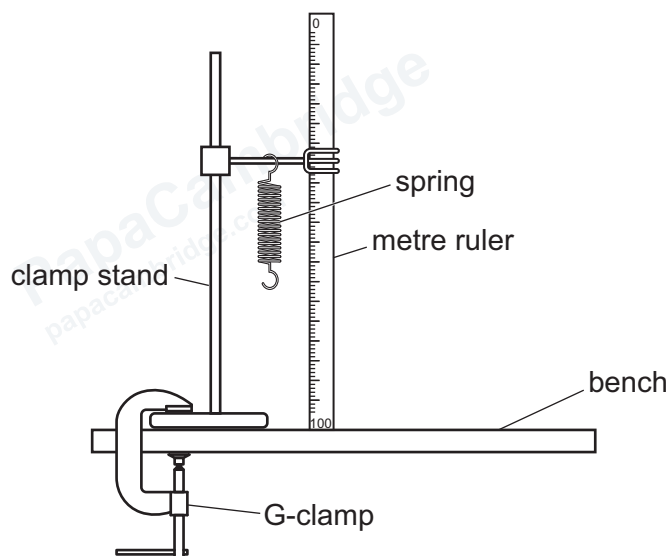


Figure 2.1

- (a) (i) Describe **two** techniques the student uses when measuring the length of the spring to ensure an accurate reading. You may draw a diagram.

1

2

[2]

(ii) Using Figure 2.2 record the initial length l_0 of the spring with no load.

Do **not** include the loops at the ends of the spring in your measurement.

Show clearly on Figure 2.2 how you made your measurement.

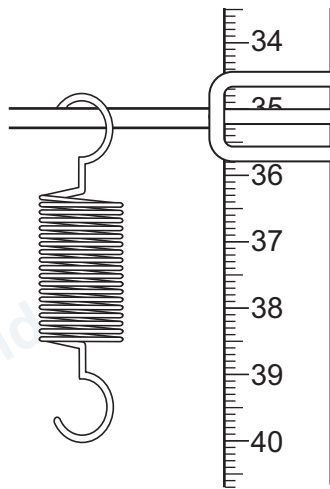


Figure 2.2

l_0 mm [2]

(b) The student:

- suspends a load $L = 1.0 \text{ N}$ from the spring
- records the new length l of the spring in Table 2.1
- calculates the extension x of the spring using the equation $x = l - l_0$
- records the value of x in Table 2.1
- repeats the procedure using loads $L = 2.0 \text{ N}$, 3.0 N , 4.0 N and 5.0 N .

Complete Table 2.1.

Table 2.1

L/N	l/mm	x/mm
1.0	42	
2.0	61	
3.0	82	
4.0	107	
5.0	125	

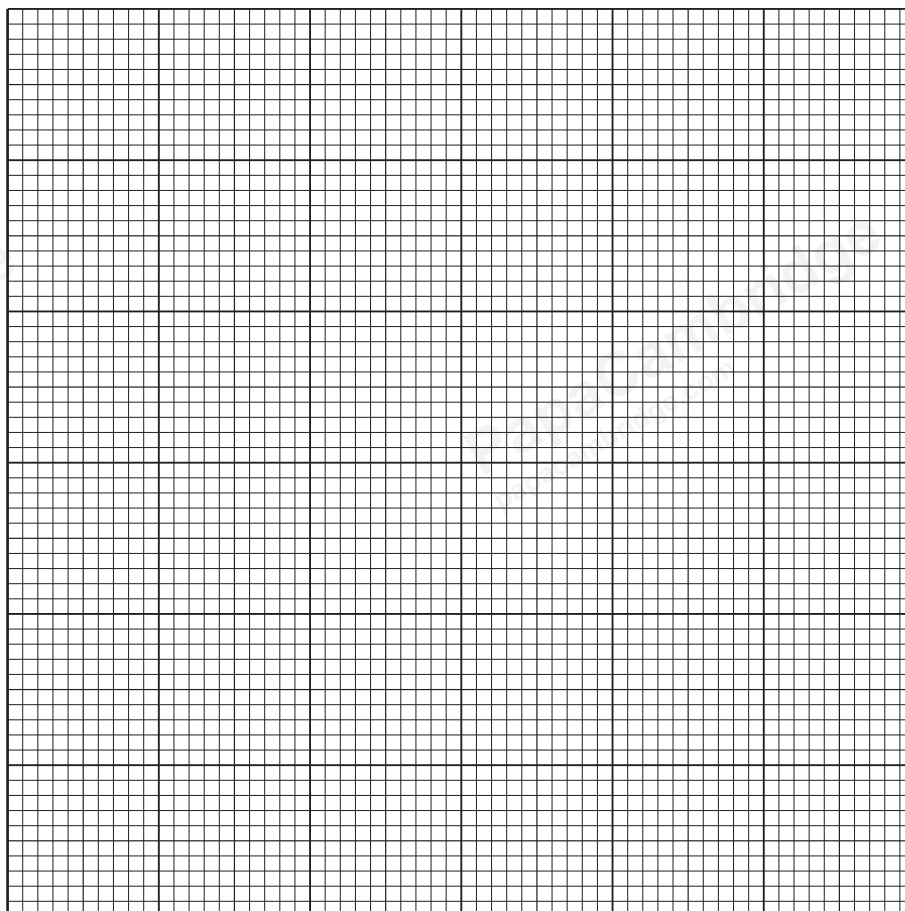
[2]



(c) (i) Plot a graph of L/N (y-axis) against x/mm (x-axis).

Start the axes from the origin (0, 0)

Draw the best-fit line.



[4]

(ii) State and explain if your line shows L and x are directly proportional.

.....

.....

..... [1]

[Total: 11]

3 A student investigates circuits containing different combinations of resistors.

(a) (i) Figure 3.1 shows circuit 1.

Circuit 1.

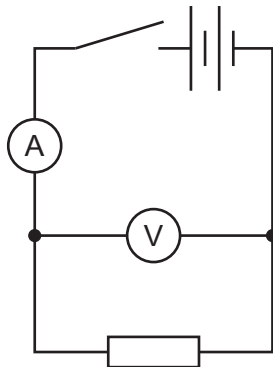


Figure 3.1

Circuit 1 contains one resistor of resistance $X \Omega$.

The student closes the switch and measures the potential difference (p.d.) and current.

Record, in Table 3.1, the voltmeter and ammeter readings shown in Figure 3.2 for circuit 1.

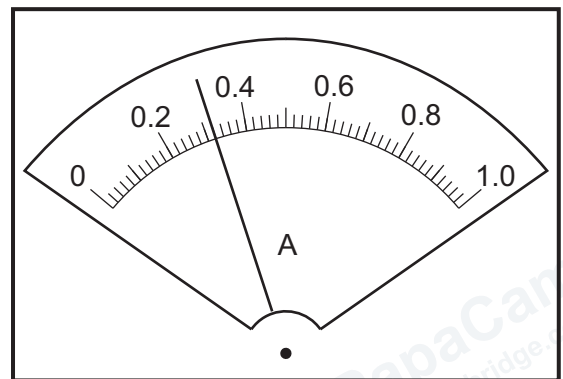
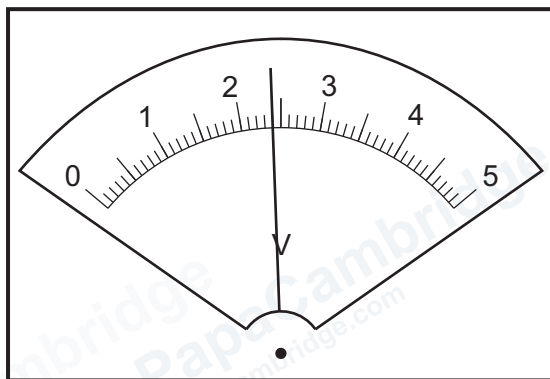


Figure 3.2

Table 3.1

circuit	V/V	I/A	R/Ω
1			
2			

[1]

(ii) Figure 3.3 shows circuit 2.

Circuit 2.

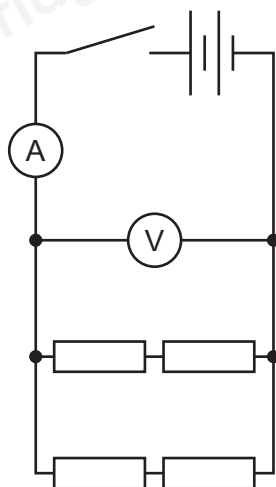


Figure 3.3

Circuit 2 contains four identical resistors each of resistance $X \Omega$.

The student closes the switch and measures the potential difference (p.d.) and current.

Record, in Table 3.1, the voltmeter and ammeter readings shown in Figure 3.4 for circuit 2.

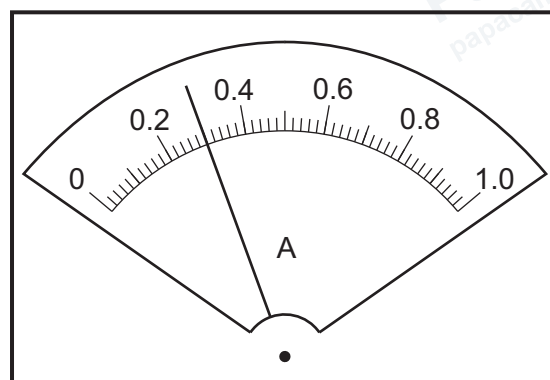
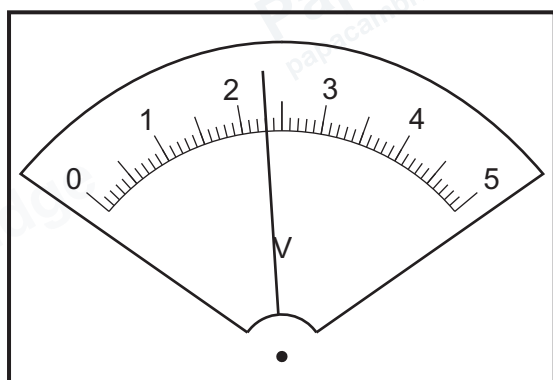


Figure 3.4

[1]



- (iii) Calculate, and record in Table 3.1, the resistance of each circuit. Give your answers to 2 significant figures.

Use your readings from Table 3.1 and the equation shown.

$$R = \frac{V}{I}$$

[3]

- (iv) Theory suggests that the resistance of circuit 1 and circuit 2 are the same.

State whether the results support this suggestion. Justify your statement by reference to values from your results.

statement

justification

.....

[2]



(v) Figure 3.5 shows circuit 3.

Circuit 3.

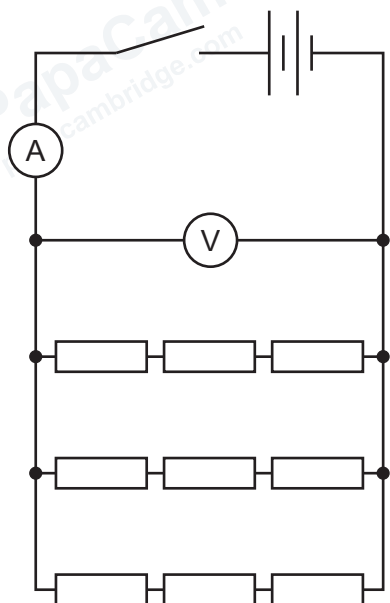


Figure 3.5

Circuit 3 contains nine identical resistors each of resistance $X \Omega$.

Predict the resistance of this combination of resistors.

..... Ω [1]

(b) Another student performs the same experiment but uses a digital ammeter.

(i) State **one** advantage of using a digital meter.

.....
 [1]

(ii) Figure 3.6 shows a digital ammeter display when it is connected in a circuit.

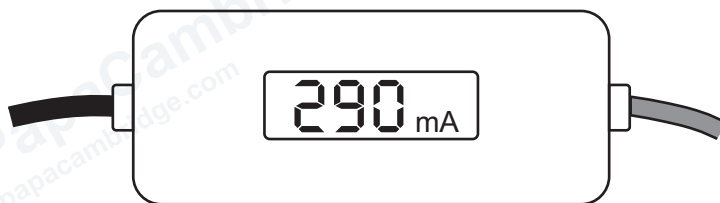


Figure 3.6

Record the value for the current shown in amperes.

..... A [1]



(c) Suggest **two** reasons why the switch is opened after readings are taken.

1

.....

2

.....

[2]

[Total: 12]





Turn over

- 4 A student investigates the rate of cooling of hot water in a container which has a lid.

Plan an experiment which enables the student to investigate the effect of different starting temperatures on the rate of cooling.

The apparatus available includes:

- a beaker
- a measuring cylinder
- a supply of hot water
- a lid

In your plan:

- state any additional apparatus required
- explain briefly how to do the investigation
- state the key variables to keep constant
- draw a table, or tables, with column headings, to show how to display your readings (you are **not** required to enter any readings in the table)
- explain how to use your readings to reach a conclusion.



[7]





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